

Notes: Hau, Massa, & Peress (RFS, 2010)

Do Demand Curves for Currencies Slope Down? Evidence from the MSCI Global Index Change

Contents

1	Big picture	2
2	Data and measurement	2
2.1	The MSCI index redefinition	2
2.2	Why December 1, 2000 is the announcement event	3
2.3	Country and stock samples	3
2.4	Weight-change variables	3
2.5	Return variables	4
2.6	FX order flow and FX trading volume	4
3	Models and empirical specifications	4
3.1	Hypothesis H1: global stock demand slopes down	4
3.2	International abnormal stock returns	4
3.3	Hypothesis H2: currency demand slopes down	5
3.4	Order-flow augmented currency regression	5
3.5	Implementation regressions	5
3.6	Currency comovement regression	5
4	Identification and empirical design	5
4.1	What identifies the main effect	5
4.2	Why the shock is plausibly uninformative	6
4.3	Why the equity result matters first	6
4.4	Addressing anticipation and leakage	6
4.5	What the paper cannot fully prove	6
5	Tables and figures	6
5.1	Figures 1–2: announcement timing and index-weight shock	7
5.2	Tables 1–2: summary statistics and stock-price effects	7
5.3	Figures 3–5: currency appreciation around announcement	8
5.4	Tables 3–6: statistical tests and currency regressions	9
5.5	Figures 6–7: FX market depth and scaling	12
5.6	Tables 7–8: implementation effects	12
5.7	Table 9 and Figure 8: permanent currency comovement	13
6	Short summary	14
7	Conclusion	15
7.1	Core conclusion	15
7.2	Economic intuition	15
7.3	Takeaway	15

1 Big picture

- **Question.** Do exogenous, uninformative demand shocks move exchange rates, and can an equity-market shock propagate into currency markets?
- **Natural experiment.** The paper studies the 2000 MSCI Global Equity Index redefinition, which moved country weights from market-cap weights toward free-float-adjusted weights.
- **Why this is useful.** The redefinition created large, mechanical portfolio rebalancing by global index funds. These flows were not caused by new country fundamentals.
- **Equity-market test.** If global demand curves for stocks slope down, upweighted stocks should rise and downweighted stocks should fall around the announcement.
- **Currency-market test.** If demand curves for currencies slope down, currencies of relatively upweighted countries should appreciate.
- **Main equity result.** Stocks that were upweighted earned positive abnormal returns. A one-standard-deviation long-short strategy across upweighted versus downweighted stocks earns about 1.18% over a twelve-day window.
- **Main currency result.** Upweighted currencies appreciated relative to downweighted currencies. Across an eight-trading-day window, the gap between the most upweighted and most downweighted currencies exceeds 2%.
- **Interpretation of price pressure.** The effect implies that even highly liquid FX markets are not perfectly elastic. Portfolio-balance effects can move macro prices.
- **Cross-asset propagation.** The demand shock originates in equities but transmits to currencies, showing that flows can connect different asset classes even without fundamental news.
- **Comovement result.** After the redefinition, upweighted currencies comove more with the basket of other MSCI currencies; downweighted currencies comove less.
- **Implementation result.** Announcement effects are stronger than implementation effects. Some effects reverse when the index changes are implemented, consistent with arbitrage and pre-positioning.
- **Economic takeaway.** Currency markets are resilient but not infinitely resilient. Large exogenous equity flows can shift exchange rates and alter the correlation structure of currencies.

2 Data and measurement

2.1 The MSCI index redefinition

- MSCI announced that it would move to a free-float-adjusted weighting methodology.
- Free float excludes holdings that are not easily available to international investors, such as government, insider, and strategic stakes.
- MSCI also raised the target market representation of its indices from roughly 60% to 85% of free-float-adjusted market capitalization.
- The redefinition changed country weights sharply. The United States, United Kingdom, and Ireland were among the main beneficiaries; Colombia, India, and the Czech Republic were among the largest percentage losers.
- The implementation occurred in two steps: 50% on November 30, 2001 and the remaining 50% on May 31, 2002.

Interpretation. The index event creates a country-level equity demand shock. Because index funds had to buy upweighted countries and sell downweighted countries, the shock is useful for identifying price pressure.

2.2 Why December 1, 2000 is the announcement event

- MSCI had discussed possible changes earlier in 2000.
- On December 1, 2000, MSCI announced that it would communicate the decision on December 10.
- On December 10, it confirmed the new methodology and implementation timetable.
- FX transaction volume peaked on the first trading day after December 1, not after December 10.
- The paper therefore treats December 1 as the relevant announcement date.

Interpretation. The relevant information appears to have entered the market around December 1. Using this date captures trading by arbitrageurs and index investors reacting to the coming redefinition.

2.3 Country and stock samples

- The MSCI ACWI contains fifty equity markets.
- The euro area countries are aggregated into a single euro-area currency unit.
- The exchange-rate analysis excludes the United States, pegged currencies such as China, Hong Kong, and Malaysia, and countries experiencing major currency crises around the event, such as Argentina, Brazil, and Turkey.
- The baseline currency sample contains thirty-three currencies.
- The stock-level analysis uses 2,436 permanent MSCI constituents, meaning stocks that belonged to the index before and after the redefinition.

Interpretation. The currency sample is designed to measure exchange-rate effects rather than crisis effects, peg mechanics, or U.S. dollar base-currency issues.

2.4 Weight-change variables

For country i , the percentage country-weight change is

$$\Delta w_i = \frac{w_i^n - w_i^o}{\frac{1}{2}(w_i^n + w_i^o)}, \quad (1)$$

where w_i^n is the new MSCI weight and w_i^o is the old MSCI weight.

The paper also scales the weight change by total FX volume and spot FX volume:

$$\Delta w_i^{FX} = \frac{w_i^n - w_i^o}{Vol_i^{FX}}, \quad \Delta w_i^{SP} = \frac{w_i^n - w_i^o}{Vol_i^{SP}}. \quad (2)$$

For stock j in country i , the stock-level weight change is

$$\Delta w_{ij} = \frac{w_{ij}^n - w_{ij}^o}{\frac{1}{2}(w_{ij}^n + w_{ij}^o)}. \quad (3)$$

Interpretation. Scaling by the average old and new weights turns the raw change into a relative demand shock. Scaling by FX volume asks whether the same index flow is bigger relative to currency-market depth.

2.5 Return variables

- Currency returns are log changes in the U.S.-dollar value of local currency. Thus $\Delta e_i > 0$ means an appreciation of currency i against the dollar.
- Stock returns are measured in both local currency and U.S. dollars.
- Event windows are symmetric six- and twelve-trading-day windows around the December 1 announcement.
- Abnormal stock returns are computed using a two-factor international asset-pricing model with domestic and global market factors.

Interpretation. The local-currency stock return isolates equity-price effects. The U.S.-dollar return combines equity and currency effects.

2.6 FX order flow and FX trading volume

- FX transaction volume around the announcement is measured with EBS and Reuters D-2000 euro-dollar spot trades.
- FX trading-volume measures come from the BIS April 2001 survey.
- The paper uses both total FX volume and spot FX volume to assess whether the price impact is related to currency-market depth.

Interpretation. The FX-volume data help calibrate the magnitude of the demand shock. If a fixed capital flow moves low-volume currencies more, that supports a downward-sloping demand curve.

3 Models and empirical specifications

3.1 Hypothesis H1: global stock demand slopes down

The first hypothesis is that stocks are imperfect substitutes globally:

$$\text{Upweighted stocks earn positive returns; downweighted stocks earn negative returns.} \quad (4)$$

The stock-level regression is

$$\Delta r_{ij} = \alpha_0 + \alpha_1 \Delta w_{ij} + \mu_{ij}. \quad (5)$$

Interpretation. Under H1, $\alpha_1 > 0$. The coefficient measures the stock-price response to a mechanical increase in index-fund demand.

3.2 International abnormal stock returns

The abnormal-return model is estimated before the announcement window:

$$r_{ijt} = \beta_{ij0} + \beta_{ijD} r_{Dit} + \beta_{ijW} r_{Wt} + \epsilon_{ijt}, \quad (6)$$

where r_{Dit} is the domestic market return and r_{Wt} is the world market return.

Interpretation. This removes ordinary domestic and global market movements so that the remaining event return is more plausibly related to MSCI reweighting.

3.3 Hypothesis H2: currency demand slopes down

The second hypothesis is that currencies are imperfect substitutes:

$$\text{Upweighted currencies appreciate; downweighted currencies depreciate.} \quad (7)$$

The baseline exchange-rate event regression is

$$\Delta e_{it} = \alpha_0 + \alpha_1 D_t + \alpha_2 (\Delta \bar{w}_i D_t) + \mu_{it}, \quad (8)$$

where D_t equals one during the event window and $\Delta \bar{w}_i$ is the demeaned scaled currency-weight change.

Interpretation. The coefficient α_2 is the key currency price-pressure coefficient. It asks whether countries with larger MSCI weight increases experience larger currency appreciations during the event.

3.4 Order-flow augmented currency regression

The daily regression adds euro-dollar orderflow and its interaction with the weight shock:

$$\Delta e_{it} = \alpha_0 + \alpha_1 D_t + \alpha_2 (\Delta \bar{w}_i D_t) + \alpha_3 (\overline{OF}_t D_t) + \alpha_4 (\Delta \bar{w}_i \overline{OF}_t D_t) + \mu_{it}. \quad (9)$$

Interpretation. Orderflow controls for broad speculative pressure in the euro-dollar market. The weight-change effect survives, which means the MSCI shock is not simply a general dollar event.

3.5 Implementation regressions

The same stock and currency regressions are applied to the first and second implementation dates:

$$\Delta r_{ij} = \alpha_0 + \alpha_1 \Delta w_{ij} + \mu_{ij}, \quad \Delta e_{it} = \alpha_0 + \alpha_1 D_t + \alpha_2 (\Delta \bar{w}_i D_t) + \mu_{it}. \quad (10)$$

Interpretation. If arbitrageurs front-run index funds at announcement, implementation effects may be weaker or reverse. This is what the paper finds in several specifications.

3.6 Currency comovement regression

The paper studies whether the MSCI event changes currency comovement:

$$\Delta e_{it} = \alpha_0 + \alpha_1 \text{Basket}_t + \alpha_2 \Delta \bar{w}_i + \alpha_3 D_t + \alpha_4 (\Delta \bar{w}_i \times \text{Basket}_t) \quad (11)$$

$$+ \alpha_5 (D_t \times \text{Basket}_t) + \alpha_6 (D_t \times \Delta \bar{w}_i) + \alpha_7 (D_t \times \Delta \bar{w}_i \times \text{Basket}_t) + \epsilon_{it}. \quad (12)$$

Interpretation. The coefficient of interest is α_7 . A positive α_7 means upweighted currencies comove more with the MSCI currency basket after the redefinition.

4 Identification and empirical design

4.1 What identifies the main effect

- The source of variation is the mechanical country reweighting generated by MSCI's methodological change.
- The event is not based on new macroeconomic information about countries.
- Country weight changes differ sharply across countries.
- The analysis compares upweighted and downweighted countries over narrow event windows.

Interpretation. The design is an event study around an exogenous portfolio-demand shock.

4.2 Why the shock is plausibly uninformative

- MSCI changed how weights were calculated, not the fundamentals of firms or countries.
- Free-float adjustment changes index representation but does not directly change cash flows.
- The currency-market effect is therefore a price-pressure effect, not a fundamentals effect.

Interpretation. This is why the paper can speak to demand-curve slopes rather than information revelation.

4.3 Why the equity result matters first

- The currency shock originates from equity rebalancing.
- Showing stock-price pressure first confirms that the MSCI redefinition generated real portfolio-demand pressure.
- The currency test then asks whether that shock propagated to FX markets.

Interpretation. H1 is not logically required for H2, but it validates the mechanism that index funds had to rebalance global equity positions.

4.4 Addressing anticipation and leakage

- MSCI had signaled a possible methodology change before December 1.
- FX volume also rose on November 30.
- The paper therefore treats estimates over narrow windows as lower bounds if some price pressure occurred before the event window.

Interpretation. Anticipation weakens the measured event effect rather than mechanically creating it, as long as the pre-event leakage is not fully captured.

4.5 What the paper cannot fully prove

- The precise magnitude of the exchange-rate elasticity is sensitive to window length and scaling choices.
- Anticipation before December 1 could move some of the effect outside the event window.
- FX market participants could have formed heterogeneous expectations about implementation timing.
- The paper studies a single major index event; the same magnitude need not apply to every flow shock.

Interpretation. The qualitative conclusion is stronger than the exact numerical price-impact estimate: large, uninformative equity flows can move exchange rates.

5 Tables and figures

This section keeps the same *Read* plus *Economic message* format. Adjacent visuals are grouped to save space and speed comparison. All visuals are directly cropped from the paper.

5.1 Figures 1–2: announcement timing and index-weight shock

Read: Figure 1:

- Excess euro-dollar spot transaction volume peaks on the first trading day after December 1, 2000.
- Volume does not show a similar peak after the December 10 confirmation.
- This supports treating December 1 as the main announcement event.

Read: Figure 2:

- Most countries are downweighted in percentage terms.
- The United States, United Kingdom, and Ireland are among the clearest upweighted countries.
- Colombia, India, and the Czech Republic are among the most downweighted.

Economic message. The first figure identifies the relevant news date. The second figure shows the cross-country demand shock used in the currency test.

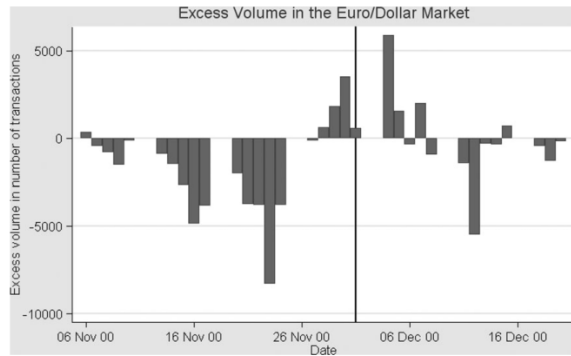


Figure 1
Plotted is the daily transaction volume in the Euro/U.S. dollar spot market around December 1, 2000, net of day-of-the-week effects
The volume data consist of the daily number of transactions in both the EBS and Reuters D-2000 spot trading platforms (see Breedon and Vitale 2004). The average trading volume (in terms of number of transactions) for the same day of the week over the six-month period from 01/08/2000 to 24/01/2001 is subtracted from the trading

(a) Euro-dollar excess trading volume

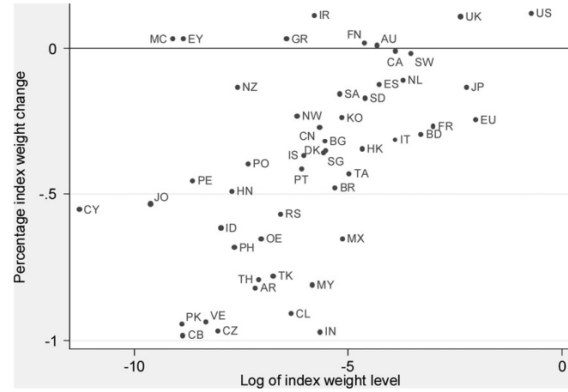


Figure 2
The percentage weight change of a country in the MSCI Global Equity Index (ACWI) is plotted against the log level of the country's weight prior to the index redefinition
The fifty index countries are Argentina (AR), Australia (AU), Brazil (BR), Canada (CA), Chile (CL), China (CN),

(b) Country weight changes in MSCI ACWI

5.2 Tables 1–2: summary statistics and stock-price effects

Read: Table 1:

- The mean percentage country-weight change is -0.4269 with a standard deviation of 0.3385.
- The country-weight changes range from -0.9825 to 0.1086.
- The six-day dollar exchange-rate return has mean 1.28% and standard deviation 2.10%.
- The equity sample has more than 2,400 stock observations.

Read: Table 2:

- The stock-level coefficient on weight change is positive in all specifications.
- For local-currency raw returns, the coefficient is 0.565 over six days and 1.303 over twelve days.
- For local-currency abnormal returns, the coefficient is 0.327 over six days and 0.502 over twelve days.
- For U.S.-dollar returns, the coefficient remains positive and significant.

Economic message. The equity evidence confirms that the MSCI event produced price pressure in global stock markets. This is the first step before testing transmission to currencies.

Table 1
Summary statistics

		Obs.	Mean	S.D.	Min	Max
Scaled weights						
Country weight change						
$\Delta w_i = \frac{w_i^p - w_i^o}{\frac{1}{2}(w_i^p + w_i^o)}$	(i) Percent (relative to mean)	33	-0.4269	0.3385	-0.9825	0.1086
$\Delta w_i^{FX} = \frac{w_i^p - w_i^o}{Vol_i^{FX}}$	(ii) Scaled by total FX volume	27	-0.2910	0.3939	-1.6812	0.0688
$\Delta w_i^{SP} = \frac{w_i^p - w_i^o}{Vol_i^{SP}}$	(iii) Scaled by FX spot volume	27	-0.6124	0.7626	-3.3807	0.2549
$\Delta w_{ij} = \frac{w_{ij}^p - w_{ij}^o}{\frac{1}{2}(w_{ij}^p + w_{ij}^o)}$	Stock weight change (percent)	2436	-0.1045	1.1778	-2.0000	2.0000
Currency market						
Δe_i^6	U.S. dollar exchange rate return	33	1.2800	2.0977	-1.7462	5.7440
Δe_i^{12}	U.S. dollar exchange rate return	33	1.2986	2.2191	-2.0783	5.4302
OF_i	Orderflow (Euro/U.S. dollar)	12	1.422	1.127	-0.334	2.658
Equity market						
Δr_{ij}^6	U.S. dollar raw stock return	2412	1.8968	78.6483	-117.7486	40.6059
	U.S. dollar abnormal stock return	2362	0.2517	7.4863	-120.7578	38.2560
Δr_{ij}^{12}	U.S. dollar raw stock return	2410	3.7697	10.7915	-98.5579	46.5046
	U.S. dollar abnormal stock return	2362	0.2435	9.8080	-107.5854	46.0544
Δr_{ij}^6	Local currency raw stock return	2412	0.83335	7.7611	-121.154	40.6059
	Local currency abnormal stock return	2362	0.0863	7.4969	-114.3833	38.2560
Δr_{ij}^{12}	Local currency raw stock return	2410	2.5729	10.7093	-102.5828	46.5046
	Local currency abnormal stock return	2362	0.1310	9.7947	-99.3108	44.6548

For the sample countries, we report summary statistics on different measures of country index reweighting, Δw_i , on U.S. dollar exchange rate return, Δe_i , and individual stock returns, Δr_{ij} , for stock j trading in country i . The country weight change $w_i^p - w_i^o$ is scaled by (i) the mean between new and old country weights $\frac{1}{2}(w_i^p + w_i^o)$, (ii) the total FX market volume in currency i , Vol_i^{FX} or (iii) the FX spot market volume, Vol_i^{SP} . In addition to spot trading volume, total FX volume includes outright forward contracts and swap contracts. Currency trading volume is available in the BIS currency market survey for April 2001 for a subsample of twenty-seven currencies. For stock returns, we report both raw and abnormal (log) returns denominated in either the local currency or U.S. dollars. Abnormal stock returns are computed for each stock using a two-factor international asset pricing model that includes both domestic and global market risk (see the text for details). Exchange rates, index, and individual stock returns are log returns in percentage points and measured over a symmetric 6- and 12-trading-day window (marked by the superscripts 6 and 12, respectively) around the announcement day of the index reweighting (December 1, 2000). The exchange rate statistics exclude Argentina, Brazil, and Turkey because they experienced a major currency crisis around the time of our analysis, as well as China, Hong Kong, and

(a) Summary statistics

Table 2
Individual stock return regression

Panel A: Individual stock returns in local currency $\Delta r_{ij} = \alpha_0 + \alpha_1 \Delta w_{ij} + \mu_{ij}$								
Window size	Raw returns OLS		Abnormal returns OLS		Raw returns OLS clustered		Abnormal returns OLS clustered	
	(1) 6 Days	(2) 12 Days	(3) 6 Days	(4) 12 Days	(5) 6 Days	(6) 12 Days	(7) 6 Days	(8) 12 Days
Constant	0.894*** (0.157)	2.716*** (0.216)	0.124 (0.154)	0.189 (0.201)	0.894** (0.427)	2.716*** (0.849)	0.124 (0.209)	0.189 (0.394)
Δw_{ij}	0.565*** (0.152)	1.303*** (0.205)	0.327** (0.150)	0.502*** (0.187)	0.565*** (0.270)	1.303** (0.544)	0.327* (0.165)	0.502** (0.243)
Obs.	2412	2410	2362	2362	2412	2410	2362	2362
R^2	0.0074	0.0206	0.0027	0.0037	0.0074	0.0206	0.0027	0.0037
Panel B: Individual stock returns in U.S. dollars								
Constant	1.960*** (0.159)	3.919*** (0.217)	0.289* (0.153)	0.303 (0.200)	0.020 (0.426)	0.039 (0.727)	0.289 (0.238)	0.303 (0.473)
Δw_{ij}	0.582*** (0.152)	1.357*** (0.204)	0.321** (0.151)	0.513*** (0.188)	0.582** (0.221)	0.136*** (0.481)	0.321** (0.158)	0.513** (0.248)
Obs.	2412	2410	2362	2362	2412	2410	2362	2362
R^2	0.0076	0.0220	0.0026	0.0037	0.0076	0.0220	0.0026	0.0038

We perform a cross-sectional least squares regression of the return of stock j in country i on the stock's percentage weight change in the MSCI Global Equity Index (ACWI), $\Delta w_{ij} = \frac{w_{ij}^p - w_{ij}^o}{\frac{1}{2}(w_{ij}^p + w_{ij}^o)}$. Panel A reports estimates based on stock returns denominated in the local currency of country i , and panel B estimates based on stock returns denominated in U.S. dollars. Raw and abnormal stock returns are measured (in percentage points) over a six- and twelve-day event window around the announcement day of the index reweighting (December 1, 2000). Abnormal returns in local currency and U.S. dollars are computed for each stock using a two-factor international asset pricing model that includes both domestic and global market risk (see the text for details). We report in parentheses robust standard errors adjusted for heteroskedasticity using White's correction ("OLS" column) and clustered by country ("OLS Clustered" column). We indicate by *, **, and *** significance at the 5%, 3%, and 1% level, respectively.

This effect is also quantitatively large. A strategy that buys an upweighted stock

(b) Individual stock return regression

5.3 Figures 3–5: currency appreciation around announcement

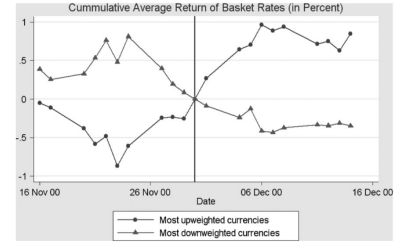
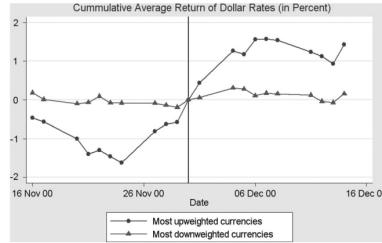
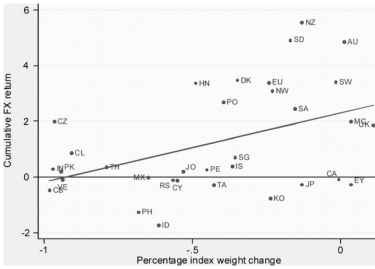
Read: Figure 3:

- Exchange-rate returns over the six-day window are positively related to country weight changes.
- The fitted line is upward sloping, indicating that upweighted currencies appreciate.
- The regression slope is about 2.49 in the paper's discussion.

Read: Figures 4–5:

- The most upweighted currencies appreciate relative to the most downweighted currencies.
- Figure 4 uses U.S. dollar exchange rates; Figure 5 uses a basket-rate benchmark.
- The gap across groups grows after the announcement and exceeds 2% over roughly eight trading days.

Economic message. The figures provide the central visual evidence that exogenous equity flows move currencies.



9

Table 3
Nonparametric test for independence of MSCI index reweighting and exchange rate change

Window size	(1) 6 Days	(2) 12 Days
Two-sided Fisher test		
p -value	0.005	0.000
One-sided Fisher test		
p -value	0.004	0.000
Number of obs.	33	33

The sample currencies are split into the sixteen most upweighted and the seventeen most downweighted currencies and alternatively into the sixteen appreciating and the seventeen most depreciating currencies. The Fisher test evaluates the likelihood of observed association under the null hypothesis that there is no relationship between the two sorting criteria. The currency appreciations are measured as the cumulative return (in U.S. dollars per local currency) over a six- and twelve-day window around the announcement day of the index reweighting (December 1, 2000). The sample countries consist of the fifty countries in the MSCI Global Equity Index (ACWI) index from which were excluded the United States, the eleven Euro zone countries that were aggregated into one, three countries with pegged currencies (China, Hong Kong, and Malaysia), and three countries that experienced a major currency crisis around the time of our analysis (Argentina, Brazil, and Turkey).

Table 4
Correlation structure of currency pairs

Groups of currency pairs	Obs.	Mean	Median	S.D.	Min	Max
(1) $Corr(\Delta e_{i1}^{up}, \Delta e_{i2}^{up})$	120	0.208	0.132	0.257	-0.093	0.985
(2) $Corr(\Delta e_{i1}^{up}, \Delta e_{i2}^{down})$	136	0.061	0.040	0.108	-0.071	0.608
(3) $Corr(\Delta e_{i1}^{down}, \Delta e_{i2}^{down})$	272	0.081	0.033	0.166	-0.281	0.862
All $Corr(\Delta e_{i1}, \Delta e_{i2})$	528	0.104	0.046	0.188	-0.280	0.985
Fisher test						
Two-sided test (p -value)	0.004					
One-sided test (p -value)	0.002					

We form all 528 currency pairs and calculate the correlation of (log) daily exchange rate changes for the four-year period from July 1, 1996, to July 1, 2000. We group currencies into the sixteen most upweighted and seventeen most downweighted ones. Reported are summary statistics for the correlation of (i) currency pairs for which both currencies are upweighted, (ii) currency pairs for which both currencies are downweighted, and (iii) currency pairs for which one currency is upweighted and the other is downweighted. The Fisher test evaluates the likelihood of the observed association (under the null hypothesis of no relationship) between the criterion (C1) of having an above median correlation for any currency pair and the criterion (C2) that both currencies in this pair are from the same group of either only up- or only downweighted currencies. The sample countries consist of the fifty countries in the MSCI Global Equity Index (ACWI) from which were excluded the United States

(a) Fisher tests and currency-pair correlations

Table 5
Regressions for weekly exchange rate returns

Scaling factor (SF_i)	(1) $\frac{1}{2}(w_i^u + w_i^d)$	(2) $\frac{1}{2}(w_i^u - w_i^d)$	(3) Vol_i^{FX}	(4) Vol_i^{FXP}
D_t	1.345*** (0.446)	1.455*** (0.508)	1.500*** (0.512)	1.500*** (0.512)
$\Delta \hat{w}_i \times D_t$	2.159*** (0.658)	2.567*** (0.802)	1.727*** (0.523)	1.957*** (0.286)
R^2 (for event week)	0.380	0.404	0.375	0.384
Currency obs.	33	27	27	27
Time obs.	174	174	174	174

We perform OLS regressions of the (log) weekly exchange rate change Δe_{it} (denominated in U.S. dollar per local currency and expressed in percentage points) on the scaled country weight changes $\Delta \hat{w}_i$, defined as the (demeaned) ratio of the weight change $w_i^u - w_i^d$ of each currency i and scaling factors (SF_i). The respective scaled weight change is interacted with a time dummy D_t , marking the event week around the announcement event (December 1, 2000), resulting in the following specification:

$$\Delta e_{it} = \alpha_0 + \alpha_1 D_t + \alpha_2 (\Delta \hat{w}_i \times D_t) + \mu_{it}.$$

As scaling factors for weight changes we use: the average of a country's new and old new weights, $\frac{1}{2}(w_i^u + w_i^d)$, in Columns 1 and 2; its total FX currency market volume, Vol_i^{FX} , in Column 3; and its total FX spot market volume, Vol_i^{FXP} , in Column (4). In addition to spot trading volume, total FX volume includes outright forward contracts and swap contracts. The estimation uses four years of weekly data exchange rate data from July 1, 1996, to July 1, 2000. The sample countries consist of the fifty countries in the MSCI Global Equity Index (ACWI) from which are excluded the United States, the eleven Euro zone countries that are aggregated into one, three countries with pegged currencies (China, Hong Kong, and Malaysia), and three countries that experienced a major currency crisis around the time of our analysis (Argentina, Brazil, and Turkey). FX volume data obtained from the BIS survey of April 2001 is available only for a reduced set of twenty-seven currencies. Standard errors for correlated panels accounting for a general covariance structure across of all currencies and time are reported in parentheses. We indicate by *, **, and *** significance at the 5%, 3%, and 1% level, respectively. The constant

(b) Weekly exchange-rate regressions

Table 6
Regression for daily exchange rate returns

Panel A: Six-day event window						
Scaling factor (SF_i)	(1) $\frac{1}{2}(w_i^n + w_i^o)$	(2) $\frac{1}{2}(w_i^n + w_i^o)$	(3) Vol_i^{FX}	(4) Vol_i^{FX}	(5) Vol_i^{SP}	(6) Vol_i^{SP}
D_t	0.230*** (0.081)	0.230*** (0.081)	0.267*** (0.094)	0.267*** (0.094)	0.267*** (0.094)	0.267*** (0.094)
$\Delta \bar{w}_i \times D_t$	0.413*** (0.130)	0.413*** (0.130)	0.286*** (0.105)	0.286*** (0.105)	0.155*** (0.057)	0.155*** (0.057)
$\overline{OF}_i \times D_t$		-0.019 (0.078)		0.075 (0.088)		0.075 (0.088)
$\Delta \bar{w}_i \times \overline{OF}_i \times D_t$		0.191 (0.123)		0.159 (0.098)		0.089 (0.053)
R^2 (event period)	0.183	0.206	0.178	0.200	0.180	0.204
Currency obs.	33	33	27	27	27	27
Time obs.	980	980	980	980	980	980
Panel B: Twelve-day event window						
Scaling factor (SF_i)	(1) $\frac{1}{2}(w_i^n + w_i^o)$	(2) $\frac{1}{2}(w_i^n + w_i^o)$	(3) Vol_i^{FX}	(4) Vol_i^{FX}	(5) Vol_i^{SP}	(6) Vol_i^{SP}
D_t	0.143** (0.057)	0.143** (0.057)	0.166** (0.067)	0.166** (0.067)	0.166** (0.067)	0.166** (0.067)
$\frac{w_i^n - w_i^o}{SF_i} \times D_t$	0.300*** (0.093)	0.300*** (0.092)	0.199*** (0.074)	0.199*** (0.074)	0.109*** (0.040)	0.109*** (0.040)
$\overline{OF}_i \times D_t$		-0.007 (0.056)		0.110 (0.063)		0.110 (0.063)
$\frac{w_i^n - w_i^o}{SF_i} \times \overline{OF}_i \times D_t$		0.229*** (0.088)		0.184*** (0.071)		0.102*** (0.038)
R^2 (event period)	0.085	0.129	0.083	0.083	0.083	0.134
Currency obs.	33	33	27	27	27	27
Time obs.	986	986	986	986	986	986

We perform OLS regressions of the (log) daily exchange rate change Δe_{it} (denominated in U.S. dollar per local currency and expressed in percentage points) on the scaled weight change $\Delta \bar{w}_i$ defined as the (demeaned) ratio of the weight change $w_i^n - w_i^o$ of each currency i and scaling factors (SF_i), the (demeaned) orderflow in the Euro/U.S. dollar market $\overline{OF}_i$, and the interaction term $\Delta \bar{w}_i \times \overline{OF}_i$. Each term is interacted with a dummy variable D_t , which marks alternatively an event window of six trading days (panel A) or twelve trading days (panel B) around the announcement of the index redefinition (December 1, 2000). Formally, the regression specification is given by

$$\Delta e_{it} = \alpha_0 + \alpha_1 D_t + \alpha_2 (\Delta \bar{w}_i \times D_t) + \alpha_3 (\overline{OF}_i \times D_t) + \alpha_4 (\Delta \bar{w}_i \times \overline{OF}_i \times D_t) + \mu_{it}.$$

As scaling factors for weight changes we use: the average of a country's new and old new weights $\frac{1}{2}(w_i^n + w_i^o)$, in Columns 1 and 2; its total FX currency market volume, Vol_i^{FX} , in Columns 3 and 4; and its total FX spot market volume, Vol_i^{SP} , in Columns 5 and 6. In addition to spot trading volume, total FX volume includes outright forward contracts and swap contracts. The estimation uses 4 years of daily data exchange rate data from July 1, 1996, to July 1, 2000. The sample countries consist of the fifty countries in the MSCI Global Equity Index (ACWI) from which we excluded the United States, the eleven Euro zone countries that were aggregated into one, three countries with pegged currencies (China, Hong Kong, and Malaysia), and three countries that experienced a major currency crisis around the time of our analysis (Argentina, Brazil, and Turkey). FX volume data obtained from the BIS survey of April 2001 is available only for a reduced set of twenty-seven currencies. Standard errors for correlated panels accounting for a general covariance structure across all currencies and time are reported in parentheses. We indicate by *, **, and *** significance at the 5%, 3%, and 1% level, respectively. The constant

Figure 5: Table 6: daily exchange-rate regressions

5.5 Figures 6–7: FX market depth and scaling

Read:

- Figure 6 plots spot FX volume against the average of old and new MSCI weights.
- Figure 7 repeats the plot for total FX trading volume.
- Both figures show that larger MSCI-weight currencies tend to have larger FX markets.
- This motivates scaling demand shocks by FX market depth.

Economic message. The currency impact is economically meaningful because the index flow is large relative to FX volume for some currencies. The price effect is not simply a country-size artifact.

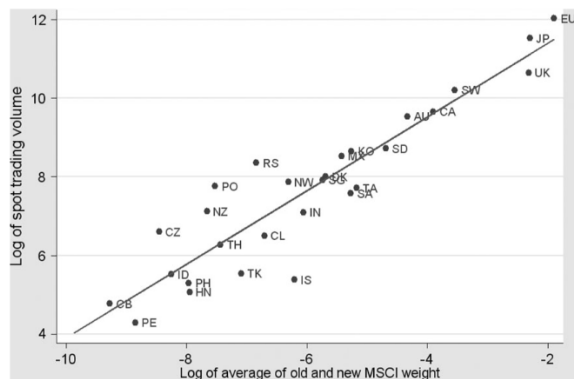


Figure 6
The spot FX trading volume as a function of the average of the old and new MSCI index weights
The figure plots the log spot FX volume against the log average of the old and new weights, $(w_t^o + w_t^o)/2$. FX volume data are obtained from the BIS April 2001 Survey. The sample countries consist of the fifty countries in the MSCI Global Equity Index (ACWI), from which were excluded the United States, the eleven Euro

(a) Spot FX volume and MSCI weights

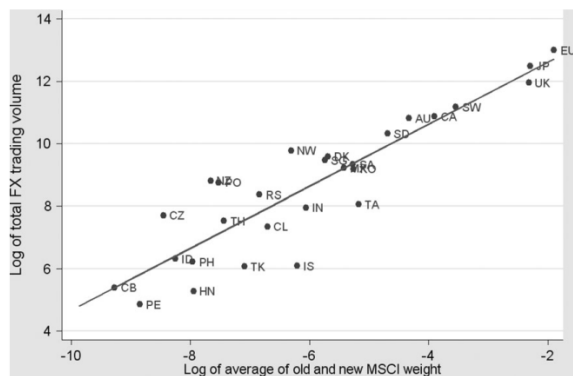


Figure 7
The spot FX trading volume as a function of the average of the old and new MSCI index weights
The figure plots the log total FX volume against the log average of the old and new weights, $(w_t^o + w_t^o)/2$. In addition to spot trading volume, Total FX Volume includes outright forward contracts and swap contracts and is obtained from the BIS April 2001 Survey. The sample countries consist of the fifty countries in the MSCI Global

(b) Total FX volume and MSCI weights

5.6 Tables 7–8: implementation effects

Read: Table 7:

- Around the first implementation date, stock reweighting coefficients are mostly negative.
- The exchange-rate tests around the first implementation date are weak and generally do not reproduce the announcement effect.
- This suggests some reversal or unwinding after the initial announcement reaction.

Read: Table 8:

- Around the second implementation date, stock coefficients remain negative in many specifications.
- Currency results differ from the announcement results, with some positive coefficients in the weekly currency regressions.
- The two implementation episodes are more complicated than the announcement episode.

Economic message. The announcement is the cleanest test of demand pressure. Implementation trading reflects anticipation, arbitrage, and unwinding, not just mechanical index demand.

Table 7
Stock and exchange returns around the first implementation of the MSCI index redefinition (November 30, 2001)

Panel A: Individual stock returns in local currency

$$\Delta r_{i,t} = \alpha_0 + \alpha_1 \Delta w_{i,t} + \mu_{i,t}$$

	Raw returns OLS		Raw returns OLS-clustered		Abnormal returns OLS clustered	
Window size	(1) 6 Days	(2) 12 Days	(3) 6 Days	(4) 12 Days	(5) 6 Days	(6) 12 Days
Constant	1.043*** (0.143)	2.240*** (0.212)	1.043 (0.740)	2.240*** (0.939)	-0.079 (0.239)	0.765** (0.329)
$\Delta w_{i,t}$	-0.453*** (0.118)	-0.864*** (0.175)	-0.453 (0.311)	-0.864*** (0.301)	-0.625*** (0.185)	-0.898*** (0.206)
Obs.	2432	2432	2432	2432	2362	2362
R^2	0.0057	0.0094	0.0057	0.0094	0.0115	0.0102

Panel B: Exchange rate returns					
Nonparametric Fisher test			Weekly currency return regressions $\Delta e_{i,t} = \alpha_0 + \alpha_1 D_t + \alpha_2 (\Delta \bar{w}_i \times D_t) + \mu_{i,t}$		
Window size	6 Days	12 Days	Window size	6 Days	12 Days
Two-sided		D_t	0.014	0.042	
p -value	0.169	1.000		(0.081)	(0.057)
One-sided			$\Delta w_i \times D_t$	-0.070	-0.050
p -value	0.112	0.571		(0.100)	(0.071)
Obs.	33	33	Currency obs.	33	33
			Time obs.	980	986

We analyze individual stock returns (panel A) and exchange returns (panel B) around the first implementation of the MSCI index redefinition. The index adjusted 50% toward the new index on November 30, 2001, and the remaining adjustment occurred on May 31, 2002. See Table 2 for details on the individual stock return regressions. See Table 3 for details on the nonparametric Fisher test carried out on exchange rates, and Table 6 for details on the regressions on daily currency returns. The currency return regressions in Panel B use the scaled (and demeaned) weight change $\Delta \bar{w}_i$ with the scaling factor $SF_i = \frac{1}{2}(w_i^t + w_i^0)$ interacted with a time dummy D_t marking alternatively a 6- or 12-trading-day event window around November 30, 2001. Standard errors for correlated panels accounting for a general covariance structure across all currencies are reported in parentheses. We indicate by *, **, and *** significance at the 5%, 3%, and 1% level, respectively.

(a) First implementation: Nov. 30, 2001

Table 8
Stock and exchange returns around the second implementation of the MSCI index redefinition (May 31, 2002)

Panel A: Individual stock returns in local currency
 $\Delta r_{i,t} = \alpha_0 + \alpha_1 \Delta w_{i,t} + \mu_{i,t}$

	Raw returns OLS		Raw returns OLS clustered		Abnormal returns OLS clustered	
	(1) 6 Days	(2) 12 Days	(3) 6 Days	(4) 12 Days	(5) 6 Days	(6) 12 Days
Constant	-1.992*** (0.114)	-3.152*** (0.176)	-1.992*** (0.408)	-3.152*** (0.667)	-0.641** (0.271)	-0.846* (0.421)
$\Delta w_{i,t}$	-0.221** (0.098)	-0.827*** (0.160)	-0.221 (0.196)	-0.827** (0.347)	-0.180 (0.116)	-0.788*** (0.197)
Obs.	2434	2434	2434	2434	2362	2362
R^2	0.0021	0.0123	0.0021	0.0123	0.0016	0.0132

Panel B: Exchange rate returns					
Nonparametric Fisher test			Weekly currency returns regressions $\Delta e_{i,t} = \alpha_0 + \alpha_1 D_t + \alpha_2 (\Delta \bar{w}_i \times D_t) + \mu_{i,t}$		
Window size	6 Days	12 Days	Window size	6 Days	12 Days
Two-sided			D_t	0.086	0.067
p -value	0.005	0.038		(0.081)	(0.057)
One-sided			$\Delta \bar{w}_i \times D_t$	0.107	0.203***
p -value	0.004	0.027		(0.100)	(0.071)
Obs.	33	33	Currency obs.	33	33
			Time obs.	980	986

We analyze individual stock returns (Panel A) and exchange returns (Panel B) around the second implementation of the MSCI index redefinition. The index adjusted the remaining 50% toward the new index on May 31, 2002 (the index adjusted the first 50% on November 30, 2001). See Table 2 for details on the individual stock return regressions. See Table 3 for details on the nonparametric Fisher test carried out on exchange rates, and Table 6 for details on the regressions on daily currency returns. The currency return regressions in Panel B use the scaled (and demeaned) weight change $\Delta \bar{w}_i$ with the scaling factor $SF_i = \frac{1}{2}(w_i^t + w_i^0)$ interacted with a time dummy D_t marking alternatively a 6- or 12-trading-day event window around May 31, 2002. Standard errors for correlated panels accounting for a general covariance structure across all currencies are reported in parentheses. We indicate by *, **, and *** significance at the 5%, 3%, and 1% level, respectively.

(b) Second implementation: May 31, 2002

5.7 Table 9 and Figure 8: permanent currency comovement

Read: Table 9:

- The triple interaction $D_t \times \Delta \bar{w}_i \times Basket_t$ is positive and statistically significant across specifications.
- In the equally weighted panel, the coefficient is about 0.903.
- In the value-weighted panel, the coefficient is about 29.522.
- The result means upweighted currencies comove more with the MSCI currency basket after the redefinition.

Read: Figure 8:

- The moving-window comovement coefficient rises around the MSCI announcement.
- The post-event average is persistently higher than the pre-event average.
- This visualizes the same comovement effect estimated in Table 9.

Economic message. The index change affects not only currency levels but also the correlation structure of currencies. A style-investing mechanism can therefore propagate from global equity benchmarks into FX comovement.

Table 9
Changes in currency comovements following the MSCI index redefinition

Panel A: Equally-weighted basket of currency returns

Error structure	(1)	(2)	(3)	(4)	(5)	(6)
	Country cluster	Day cluster	Both clusters	Prais-Winsten	Country FE	
$Basket_t$	0.636***	0.636***	0.636***	0.636***	0.636***	0.636***
$\Delta \bar{w}_i (\times 10^{-1})$	-0.004**	-0.004*	-0.004*	-0.004*	-0.004*	-0.004
$D_t (\times 10^{-2})$	-0.012	-0.012	-0.012**	-0.012	-0.012***	-0.012
$\Delta \bar{w}_i \times Basket_t$	0.117*	0.117	0.117	0.117	0.117*	0.117*
$D_t \times Basket_t$	0.068	0.276	0.095	0.283	0.068	0.068
$D_t \times \Delta \bar{w}_i (\times 10^{-2})$	0.223***	0.223	0.223***	0.223	0.223***	0.223***
$D_t \times \Delta \bar{w}_i \times Basket_t$	0.038	0.134	0.059	0.141	0.014	0.038
Currency obs.	33	33	33	33	33	33
Time obs.	1044	1044	1044	1044	1044	1044
R^2	0.059	0.059	0.059	0.059	0.059	0.059

Panel B: Value-weighted basket of currency returns

$Basket_t$	31.580***	31.580***	31.580***	31.580***	31.580***	31.580***
$\Delta \bar{w}_i (\times 10^{-1})$	2.064	5.117	2.310	5.221	0.952	2.063
$D_t (\times 10^{-2})$	-0.004**	-0.004**	-0.004**	-0.004	-0.004**	-0.004
$\Delta \bar{w}_i \times Basket_t$	0.002	0.002	0.002	0.000	0.002	0.000
$D_t \times Basket_t$	-0.025***	-0.025***	-0.025**	-0.025**	-0.025***	-0.025***
$D_t \times \Delta \bar{w}_i (\times 10^{-2})$	0.009	0.011	0.011	0.012	0.008	0.009
$D_t \times \Delta \bar{w}_i \times Basket_t$	25.338***	25.338	25.338***	25.338*	25.338***	25.338***
Currency obs.	35.850	15.030	4.259	15.145	3.022	3.842
Time obs.	10.261***	10.261**	10.261***	10.261**	10.261***	10.261***
R^2	2.250	4.627	2.476	4.741	1.291	2.250
	0.0268	0.0268	0.0268	0.0268	0.0268	0.0268
	0.0252	0.0266	0.0256	0.0269	0.0244	0.0252
	29.522***	29.522**	29.522***	29.522***	29.522***	29.522***
	5.015	11.148	5.123	11.197	4.158	5.007
Currency obs.	33	33	33	33	33	33
Time obs.	1044	1044	1044	1044	1044	1044
R^2	0.058	0.058	0.058	0.058	0.058	0.058

This table examines whether currencies that are upweighted (downweighted) comove more (less) with the other currencies in the index. Daily data on exchange rates covering two-year periods before and after the announcement event (1998-1999 and 2003-2004) are used to estimate the following panel regression:

$$\Delta e_{it} = \alpha_0 + \alpha_1 Basket_t + \alpha_2 \Delta \bar{w}_i + \alpha_3 D_t + \alpha_4 (\Delta \bar{w}_i \times Basket_t) + \alpha_5 (D_t \times Basket_t) + \alpha_6 (D_t \times \Delta \bar{w}_i) + \alpha_7 (D_t \times \Delta \bar{w}_i \times Basket_t) + \epsilon_{it},$$

where Δe_{it} measures the percentage change in the currency i on day t , $Basket_t$ is the return on a constructed index of currencies on day t (equally or value weighted), $\Delta \bar{w}_i$ represents the scaled weight change defined as the (demeaned) ratio of the weight change $w_t^i - w_0^i$ of currency i and the scaling factor $SF_i = \frac{1}{2}(w_t^i + w_0^i)$, D_t is a dummy variable that equals 0 in the pre-event period (1998-1999) and 1 in the post-event period (2003-2004). Panel A shows the results based on the basket constructed as an equally weighted index of currency returns, while Panel B presents the results based on the basket constructed as a value-weighted index of currency returns. We report OLS coefficients for the pooled currency data and consider alternative specification based on different treatments of the errors. Column (1) presents standard errors corrected for heteroskedasticity using White's adjustment. Columns (2), (3), and (4) allow for additional error clustering by country, day and country, and day, respectively. In Column (5), the coefficients are estimated using a Prais-Winsten regression, which

(a) Currency comovement regressions

the date of the MSCI index change on December 1, 2000. The coefficient $\hat{\alpha}_3$ is persistently higher for the two-year period following the index change.

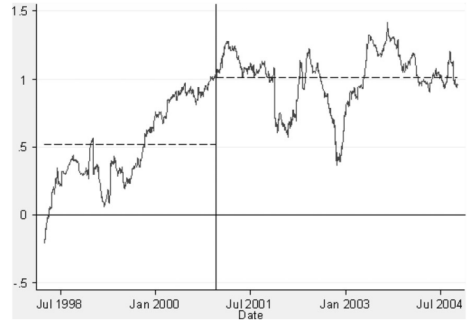


Figure 8
The graph plots the OLS regression coefficient $\hat{\alpha}_3$ of pooled exchange rate returns Δe_{it} on the interaction term $\Delta \bar{w}_i \times Basket_t$ for a moving window of 180 days. The regression controls for the index percentage weight change of a country $\Delta \bar{w}_i$ and the return $Basket_t$ of the currency basket. The comovement coefficient is plotted for the mid-point of the time window with equal weight

(b) Moving-window comovement coefficient

6 Short summary

- The paper asks whether currency demand curves slope down.
- The MSCI Global Equity Index redefinition provides a clean demand shock because it mechanically changes country weights.
- The relevant announcement date is December 1, 2000, as shown by euro-dollar transaction volume.
- The equity sample contains about 2,436 permanent index constituents.
- The baseline currency sample contains thirty-three currencies after removing the United States, pegged currencies, euro-area duplicates, and crisis countries.
- Upweighted stocks earn positive returns around the announcement.
- Upweighted currencies appreciate relative to downweighted currencies.
- The gap between the most upweighted and most downweighted currencies exceeds 2% over about eight trading days.

- The currency result survives nonparametric tests, weekly regressions, daily regressions, and alternative scalings by FX volume.
- Orderflow controls do not explain away the weight-change effect.
- Implementation effects are more mixed, consistent with pre-positioning and reversal.
- After the redefinition, upweighted currencies comove more with the other MSCI currencies.
- The results support portfolio-balance theory: currencies are imperfect substitutes, so exogenous capital flows can move exchange rates.

7 Conclusion

7.1 Core conclusion

Hau, Massa, and Peress provide direct event-study evidence that demand curves for currencies slope down. A mechanical redefinition of the MSCI Global Equity Index generated exogenous equity flows, and countries whose equity-market representation rose experienced currency appreciation.

7.2 Economic intuition

Index funds had to buy more equity in upweighted countries and less equity in downweighted countries. These equity purchases required currency demand. If investors are not perfectly willing to absorb currency imbalances at unchanged exchange rates, the currencies of upweighted countries must appreciate. The event therefore shows that equity flows can create exchange-rate price pressure.

7.3 Takeaway

The paper links equity index rebalancing, currency price pressure, and currency comovement. Its central lesson is that macro prices such as exchange rates can be moved by institutional demand shocks that contain little or no fundamental information.